

# Miles Smith

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## Education

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|----------------|--------------------------------------------------------------------------------------|---------------|
| 2023 - Present | Massachusetts Institute of Technology<br><i>SM/Ph.D., Mechanical Engineering</i>     | Cambridge, MA |
| 2020 - 2022    | Stanford University<br><i>MS, Atmosphere and Energy (CEE)</i>                        | Stanford, CA  |
| 2016 - 2020    | University of Maryland, Baltimore County (UMBC)<br><i>BS, Mechanical Engineering</i> | Baltimore, MD |

## Experience

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| 2023 - Present | Graduate Research Assistant<br><i>Massachusetts Institute of Technology, Department of Mechanical Engineering</i><br><i>Advisor: Brian Anthony</i>                                  |
|                | Research in physiology and biomechanics.                                                                                                                                            |
| 2022 - 2023    | Mechanical Engineer (Fellowship)<br><i>Otherlab / Channing Street Copper</i>                                                                                                        |
|                | Developed embedded system firmware and integrated control logic for pre-seed stage start-up company building battery-powered induction stoves.                                      |
| 2021 - 2022    | Energy Systems Group Summer Research Program<br><i>Lincoln Laboratory, Massachusetts Institute of Technology</i><br><i>Supervisors: Erik Limpaecher, Theodore Bloomstein, Ph.D.</i> |
|                | Developed a control system using Arduino to characterize the battery state-of-health as a function of the battery's internal resistance.                                            |

## Honors and Awards

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| 2024        | Wunsch Foundation Silent Hoist and Crane Award (MIT)                                                                                                 |
| 2021- 2022  | GEM Full Fellowship (Stanford)                                                                                                                       |
| 2021        | USTFCCCA NCAA Division I Men's Track & Field All-Academic Award (Stanford)                                                                           |
| 2020        | MDCAAA Postgraduate Scholarship Award Winner (UMBC)                                                                                                  |
| 2020        | Arthur Ashe Jr. Sports Scholar Award (UMBC)                                                                                                          |
| 2020        | Earnestine Bailey Baker Scholar (UMBC)                                                                                                               |
| 2018 - 2019 | Undergraduate Research Award (UMBC)<br><i>Project: Piezoelectric Energy Harvesting Research For Self-Sustainable Wind Turbine Monitoring Systems</i> |

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|-------------|---------------------------------------------|
| 2016 - 2020 | Meyerhoff Premier Scholarship Award (UMBC)  |
| 2016 - 2020 | Track and Field Athletic Scholarship (UMBC) |

### Journal Publications

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1. Jacobson, M.Z.; von Krauland, A.-K.; Coughlin, S.J.; Palmer, F.; **Smith, M.**, Zero Air Pollution and Zero Carbon From All Energy at Low Cost and Without Blackouts in Variable Weather Throughout the U.S. With 100% Wind-Water-Solar and Storage. *Renewable Energy* 2022.
  2. Jacobson, M.Z.; von Krauland, A.-K.; Burton, Z.F.; Coughlin, S.J.; Jaeggli, C.; Nelli, D.; Nelson, A.J.H.; Shu, Y.; **Smith, M.**; Tan, C.; Wood, C.D.; Wood, K.D. Transitioning All Energy in 74 Metropolitan Areas, Including 30 Megacities, to 100% Clean and Renewable Wind, Water, and Sunlight (WWS). *Energies* 2020, 13, 4934.

### Patents

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1. Induction Heating Adapter System and Method (Provisional)
  2. Battery-integrated appliance system and method (Provisional)
  3. Battery-integrated water heater system and method (Provisional)

### Conference Proceedings

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1. Jung, H.J.; Chervin, S.; **Smith, M.**; Lee, S. Design of an impact-driven piezoelectric energy harvester with gravity-induced rotator for wind turbine blade monitoring system (Conference Presentation). Proc. SPIE 10595, Active and Passive Smart Structures and Integrated Systems XII, 105951Y (3 April 2018).

### Affiliations

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|-------------|------------------------------------------------------------------|
| 2022 - 2024 | Semi-Professional Athlete at Peninsula Distance Club             |
| 2020 - 2021 | Stanford Track and Field Team                                    |
| 2016 - 2020 | UMBC Meyerhoff Scholar                                           |
| 2016 - 2020 | UMBC Cross Country and Track & Field Team (Captain)              |
| 2019-2020   | Tau Beta Pi Engineering Honor Society                            |
| 2018 - 2020 | Chi Alpha Sigma Student-Athlete Honor Society                    |
| 2018        | Louis Stokes Alliance for Minority Participation (LSAMP) Scholar |